

Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

Project Work Plan

EasyStego: A Drag-and-Drop Image Steganography Tool

Course Name: Software Development I
Course Code: CSE 2216

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Date of Submission: 21-Jun-2026

10-Week Work Plan for EasyStego Project

10-Week Overview

Week	Phase
1	Project Planning & Concept Finalization
2	Requirements Analysis & System Design
3	Environment Setup & Project Initialization
4	GUI Layout: Tabs, Drop Zones & Theming
5	Drag-and-Drop Integration
6	Core Logic: LSB Embedding & Capacity Calculation
7	Core Logic: Message Extraction & Decoding
8	Special Features & Polish
9	Error Handling & Testing
10	Final Testing, Documentation & Submission

Week 1: Project Planning & Concept Finalization

- Select project topic and objectives.
- Define steganography tool features and scope.
- Prepare initial project plan.

Week 2: Requirements Analysis & System Design

- Gather functional requirements for embedding, extraction, and capacity calculation.
- Design system workflow: Hide Message and Reveal Message flows.
- Create basic wireframes for the two-tab interface and drop zones.
- Draft the initial README.md documentation outlining project purpose and usage.

Week 3: Environment Setup & Project Initialization

- Install JDK 11+ and configure Visual Studio Code.
- Set up the Java Swing development environment.
- Create the project structure and the main EasyStego class extending JFrame.

Week 4: GUI Layout: Tabs, Drop Zones & Theming

- Build the JTabbedPane with Hide Message and Reveal Message tabs.
- Implement the dashed-border drop zone label and capacity display.
- Apply the Segoe UI font, Windows-blue accent color, and styled buttons.

Week 5: Drag-and-Drop Integration

- Implement the custom ImageTransferHandler class extending TransferHandler.
- Support DataFlavor.javaFileListFlavor to accept image files dropped from the OS.
- Render scaled image previews inside the drop zone on successful drop.

Week 6: Core Logic: LSB Embedding & Capacity Calculation

- Implement embedMessage() to write message bits into pixel LSBs with the ####END### terminator.
- Implement calculateCapacity() to compute and display (width × height) / 8 in real time.
- Wire the Encrypt & Save button to embedMessage() and a JFileChooser save dialog.

Week 7: Core Logic: Message Extraction & Decoding

- Implement extractMessage() to read pixel LSBs sequentially and reconstruct bytes.
- Detect the ####END### terminator and decode bytes back into readable text.
- Trigger automatic decoding when an image is dropped into the Reveal tab.

Week 8: Special Features & Enhancement

- Add the Copy Text to Clipboard button for decoded messages.
- Refine drop-zone visual feedback (border highlight, icon preview) after a successful load.
- Improve overall user experience and message-area styling.

Week 9: Error Handling & Testing

- Handle missing image and empty message cases with JOptionPane error dialogs.
- Test embedding and extraction across PNG images of varying dimensions.
- Fix bugs identified during testing and optimize pixel-loop performance.

Week 10: Final Testing & Documentation

- Perform complete system testing of both Hide and Reveal workflows.
- Finalize project documentation and capture screenshots.
- Finalize and submit the EasyStego project.